

Donghae Seo

Postdoctoral Researcher / KAIST

donghaeseo.github.io | [Google Scholar](#) | [ORCID: 0009-0005-3455-2430](#)

Current Position

Postdoctoral Researcher

Sep 2026 – Present

Natural Science Research Institute, Korea Advanced Institute of Science and Technology (KAIST)
Daejeon, South Korea | Alternative Military Service (Expected completion: Aug 2027)

Education

Ph.D. in Physics

Mar 2021 – Aug 2026

Pohang University of Science and Technology (POSTECH), Pohang, South Korea
Advisors: Prof. Gil Young Cho (Co-advised by Prof. Hee-Cheol Kim)
Thesis: *Classification of Topological Orders and Its Applications*

B.S. in Physics

Mar 2017 – Feb 2021

Pohang University of Science and Technology (POSTECH), Pohang, South Korea

Research Experience

Visiting Graduate Researcher

Jun 2024 – Jul 2024

Cavendish Laboratory, University of Cambridge, Cambridge, UK
Host: Prof. Robert-Jan Slager

Conducted collaborative research on noninvertible duality [Physical Review B **111**, 115112 (2025)].

Publications & Preprints

* Equal contributions † Corresponding author

Preprints

- [8] Wavefunctions for Anyon Superconductors
D. Seo*, T. Lee*, and G. Y. Cho†, arXiv:2609.04187
- [7] A Unified Categorical Description of Quantum Hall Hierarchy and Anyon Superconductivity
D. Seo, T. Lee, G. Y. Cho†, arXiv:2602.03848

Published and Accepted Papers

- [6] U(2) Chern-Simons-Ginzburg-Landau theory of fractional quantum Hall hierarchies
T. Lee, G. Y. Cho†, D. Seo†, Physical Review B, accepted as a Letter (2026)
- [5] Superconductivity proximate to non-Abelian fractional spin Hall insulator in twisted bilayer MoTe₂
C.-E. Ahn*, D. Seo*, G. Lee, Y. Kim, G. Y. Cho†, Nature Communications **17**, 7005 (2026)
- [4] Most Two-Dimensional Bosonic Topological Orders Forbid Sign-Problem-Free Quantum Monte Carlo Simulation: Nonpositive Gauss Sum as an Indicator
D. Seo, M. You, H.-C. Kim†, and G. Y. Cho†, Physical Review Letters **135**, 146602 (2025)
- [3] Noninvertible duality and symmetry topological order of one-dimensional lattice models with spatially modulated symmetry
D. Seo, G. Y. Cho†, and R.-J. Slager, Physical Review B **111**, 115112 (2025)
- [2] Modular extension of topological orders from congruence representations
D. Seo*, M. You*, G. Y. Cho†, and H.-C. Kim†, Physical Review B **109**, 155105 (2024)
- [1] Classification of fermionic topological orders from congruence representations
G. Y. Cho†, H.-C. Kim†, D. Seo†, M. You†, Physical Review B **108**, 115103 (2023)

Talks & Presentations

Seminars

- [7] *A Unified Approach to Quantum Hall Hierarchy and Anyon Superconductivity*
Condensed Matter Theory Group, Massachusetts Institute of Technology Jul 2026
- [6] *A Unified Approach to Quantum Hall Hierarchy and Anyon Superconductivity*
Department of Physics, Yale University Jul 2026
- [5] *Classification of Topological Orders and Its Applications*
Cavendish Laboratory, University of Cambridge Jun 2024

Conference Presentations

- [4] *Classification of Fermionic Topological Orders using Congruence Representations*
2023 KPS Spring Meeting, Daejeon, South Korea Apr 2023
Outstanding Presentation Award
- [3] *Classification of Fermionic Topological Orders from Congruence Representations*
APS March Meeting 2023, Las Vegas, NV, USA Mar 2023
- [2] *Algebraic Theory of Anyons*
Frontiers of Theoretical Physics 2022, Seogwipo, South Korea Nov 2022
- [1] *Classification of Non-Unitary MTC and Topological Phases*
APS March Meeting 2022, Chicago, IL, USA Mar 2022

Awards & Fellowships

- Byung Il Min Solid State Theory Paper Award, POSTECH** 2026
One recipient per year, selected from among all graduate students and postdoctoral researchers
- Outstanding Paper Award, POSTECH** 2026
Awarded for the year's top graduate student publications
- POSTECHIAN Fellowship, POSTECH** 2025
Highly competitive university-wide fellowship awarded to a select number of students
- Outstanding Presentation Award, The Korean Physical Society** 2023
Awarded for an outstanding oral presentation at the 2023 KPS Spring Meeting
- BK21 Interdisciplinary Scholar Fellowship, POSTECH** 2021–2024
One recipient in the annual physics cohort; provides a full stipend

Teaching & Mentoring

- Undergraduate Research Supervision** 2025 – 2026
Directed Taegon Lee's research on field theory of quantum Hall hierarchies, resulting in a first-author paper accepted as a Letter in Physical Review B on which I served as corresponding author
- Teaching Assistant, Department of Physics, POSTECH** 2021 – 2022
Introduction to Relativity; Physics Laboratory I; Quantum Mechanics I
- Undergraduate Peer Mentor, POSTECH** 2019 – 2020
General Physics I, General Physics II
- Residential Advisor, POSTECH** 2019 – 2020

Academic Service

- Referee** *SciPost Physics* and *Physical Review B*
- Organizer** Condensed Matter Physics Journal Club at POSTECH